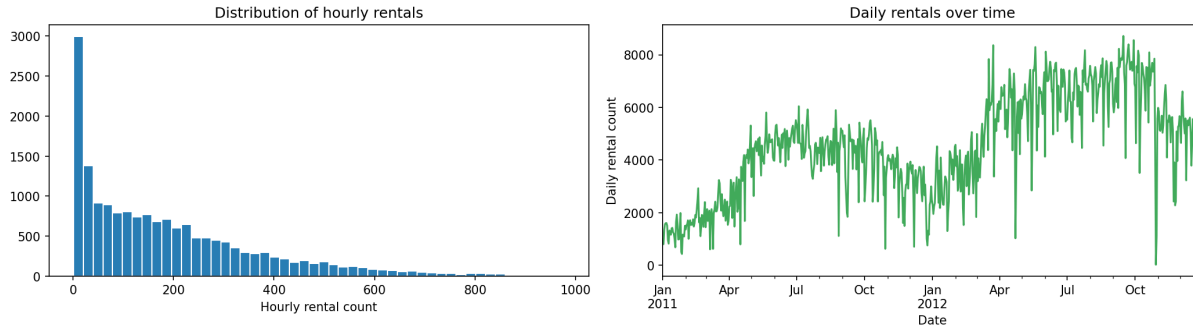


Bike Sharing Demand Forecasting

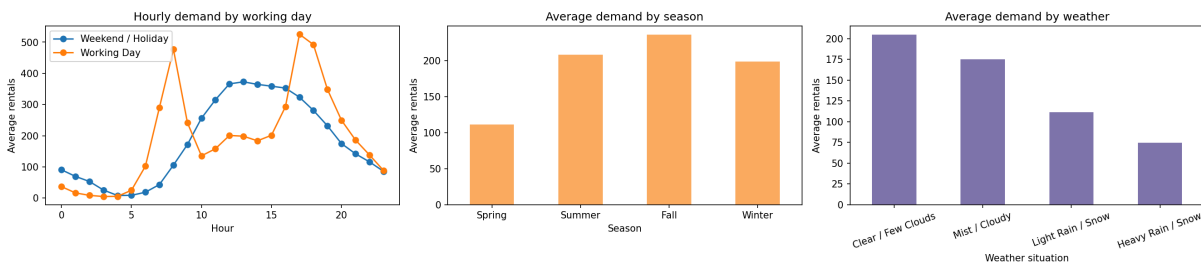
Additional Plot Pages

Demand Trend



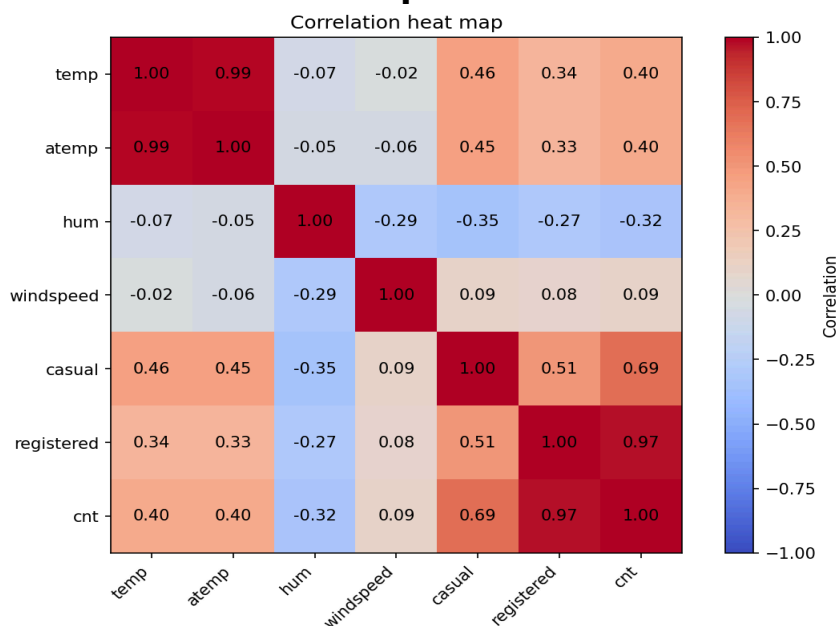
Distribution of hourly rentals and daily total demand over time.

Season And Weather Demand



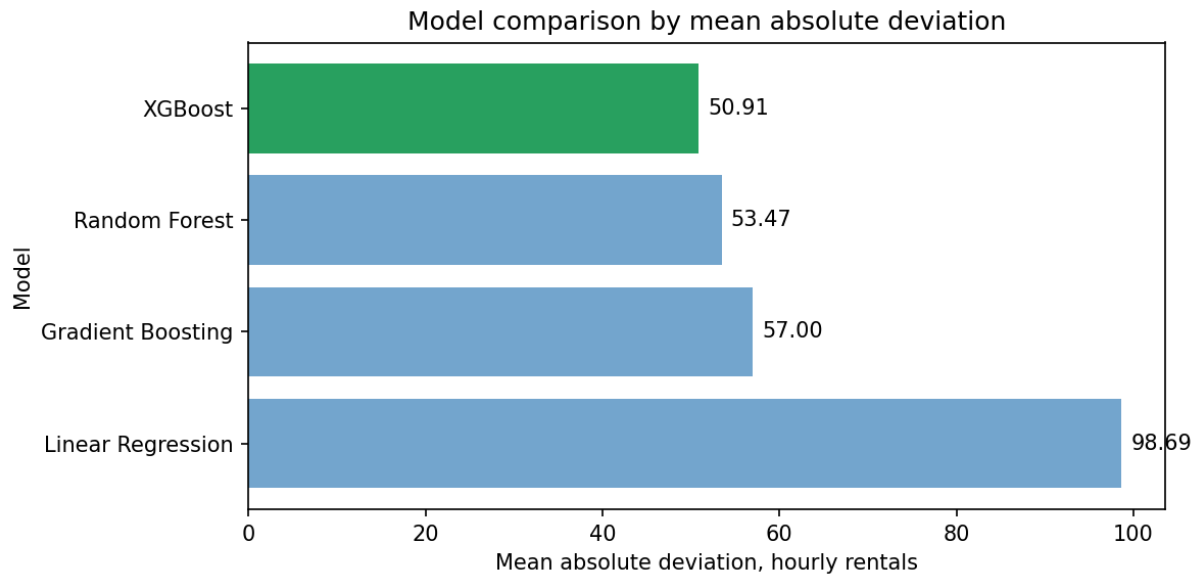
Average demand by working day, season, and weather condition.

Correlation Heat Map



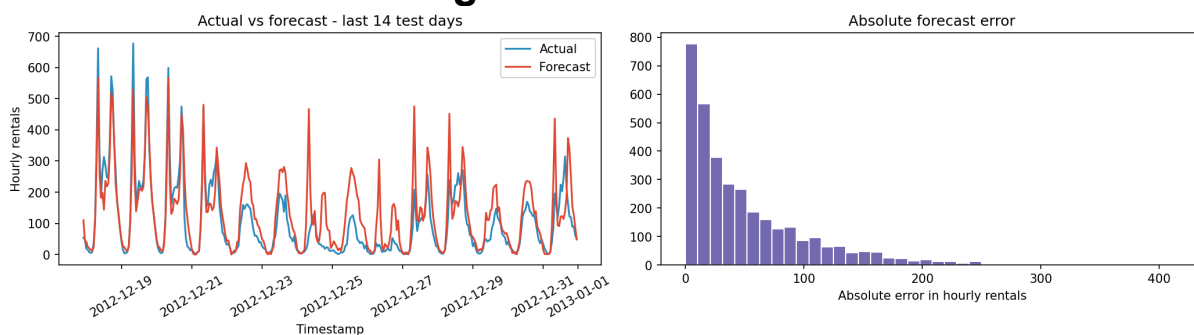
Relationships among weather variables and demand components.

Model Selection Evidence



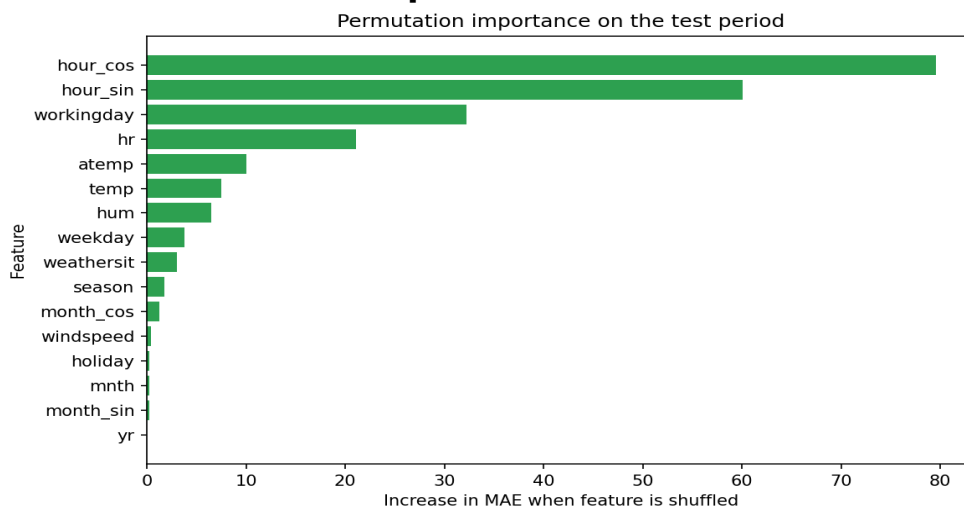
The supporting workflow compares candidate models; XGBoost has the lowest mean absolute deviation.

XGBoost Forecast Diagnostics



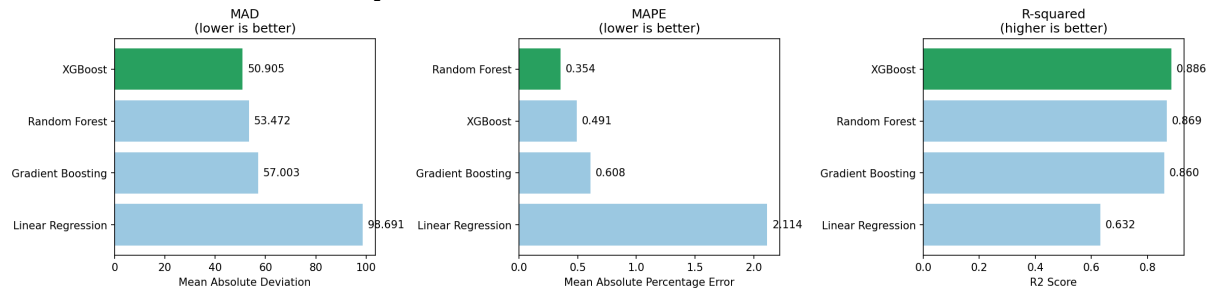
Actual vs forecast for the final test period and absolute forecast error distribution.

XGBoost Feature Importance



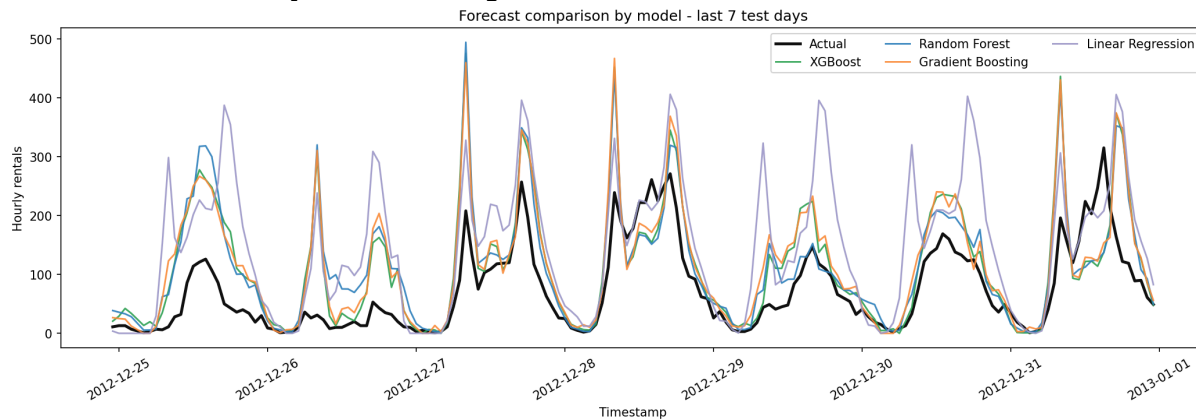
Permutation importance showing which inputs most affect forecast accuracy.

Model Metric Comparison



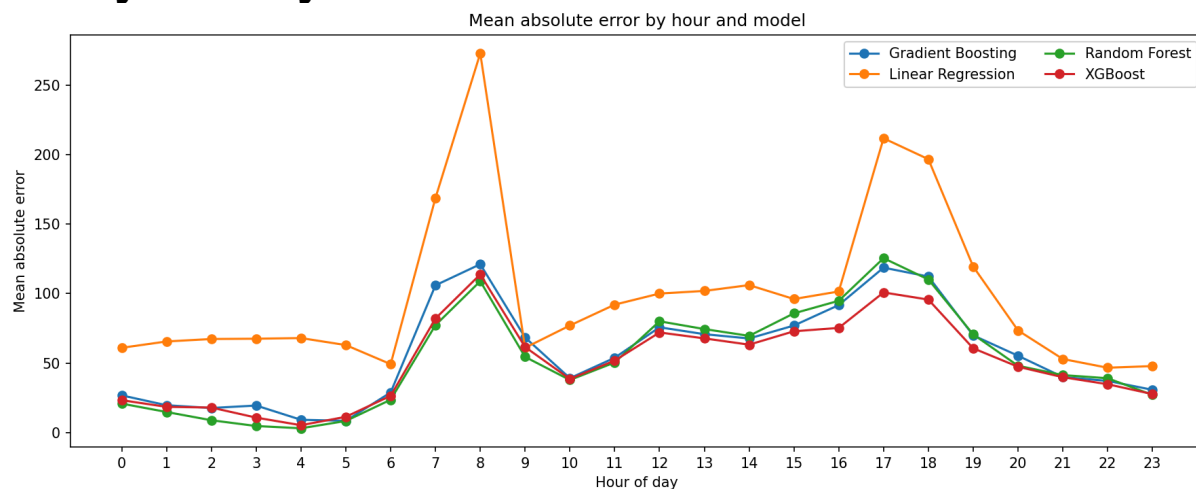
Comparison across MAD, MAPE, and R-squared.

Forecast Comparison By Model



Actual demand compared with each model's forecast over the latest test-period window.

Hourly Error By Model



Mean absolute error by hour of day for each trained model.